



CS6660: MATHEMATICAL FOUNDATIONS OF DATA SCIENCE (PROBABILITY)

QUIZ 01

DATE: 12 SEPTEMBER 2026

1. (3 Marks)

In the lectures, we saw that if A, B are two events with $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.

Construct an example with two events A, B such that A is a **strict subset** of B and $\mathbb{P}(A) = \mathbb{P}(B)$.

Clearly indicate $(\Omega, \mathcal{F}, \mathbb{P})$ in your example, and justify your steps clearly.



2. (2 Marks)

Let Ω be a sample space, and let $\mathcal{F}$ be a σ -algebra of subsets of Ω . Fix $B \in \mathcal{F}$, and consider the collection

$$\mathcal{G} = \{A \cap B : A \in \mathcal{F}\}.$$

That is, $\mathcal{G}$ is a collection of subsets of B formed by taking the intersection of each set in $\mathcal{F}$ with B . Show that $\mathcal{G}$ is a σ -algebra of subsets of B .